

Figure 421. Message buffer structure

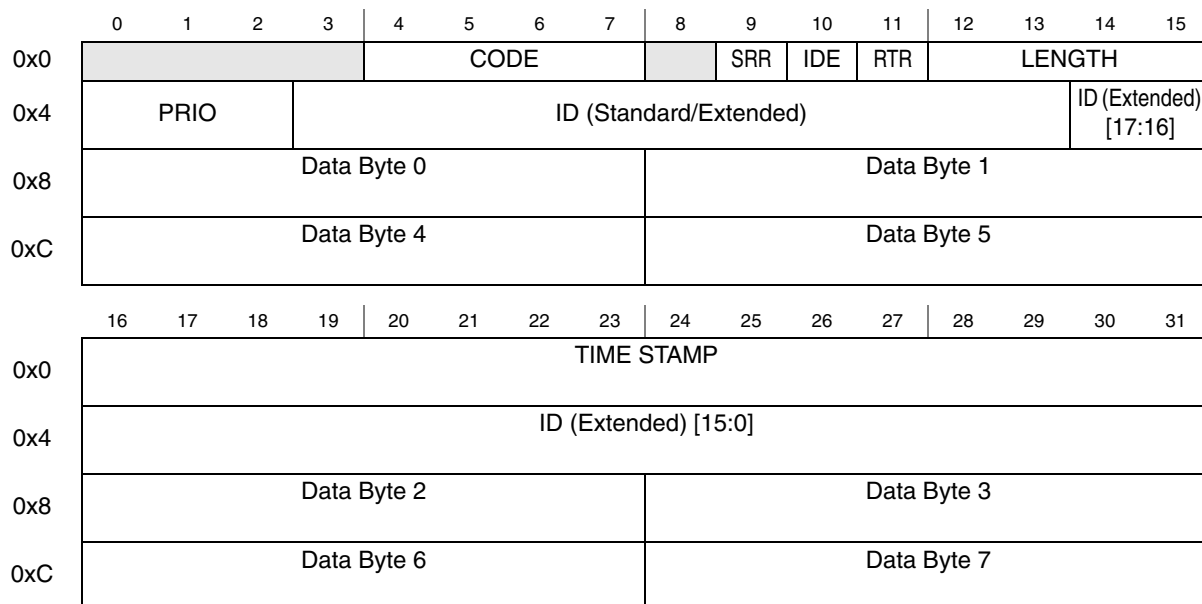


Table 398. Message Buffer structure field description

Field	Description
CODE	Message Buffer Code This 4-bit field can be accessed (read or write) by the CPU and by the Flexcan module itself, as part of the message buffer matching and arbitration process. The encoding is shown in Table 399 and Table 400. See Section 21.4 Functional description for additional information.
SRR	Substitute Remote Request Fixed recessive bit, used only in extended format. It must be set to 1 by the user for transmission (Tx Buffers) and will be stored with the value received on the CAN bus for Rx receiving buffers. It can be received as either recessive or dominant. If FlexCAN receives this bit as dominant, then it is interpreted as arbitration loss. 0 Dominant is not a valid value for transmission in Extended Format frames. 1 Recessive value is compulsory for transmission in Extended Format frames.
IDE	ID Extended Bit This bit identifies whether the frame format is standard or extended. 0 Frame format is standard 1 Frame format is extended
RTR	Remote Transmission Request This bit is used for requesting transmissions of a data frame. If FlexCAN transmits this bit as 1 (recessive) and receives it as 0 (dominant), it is interpreted as arbitration loss. If this bit is transmitted as 0 (dominant), then if it is received as 1 (recessive), the FlexCAN module treats it as bit error. If the value received matches the value transmitted, it is considered as a successful bit transmission. 0 Indicates the current MB has a Data Frame to be transmitted 1 Indicates the current MB has a Remote Frame to be transmitted

Table 398. Message Buffer structure field description (continued)

Field	Description
LENGTH	Length of Data in Bytes This 4-bit field is the length (in bytes) of the Rx or Tx data, which is located in offset 0x8 through 0xF of the MB space (see Table 421). In reception, this field is written by the FlexCAN module, copied from the DLC (Data Length Code) field of the received frame. In transmission, this field is written by the CPU and corresponds to the DLC field value of the frame to be transmitted. When RTR=1, the Frame to be transmitted is a Remote Frame and does not include the data field, regardless of the Length field.
TIME STAMP	Free-Running Counter Time Stamp This 16-bit field is a copy of the Free-Running Timer, captured for Tx and Rx frames at the time when the beginning of the Identifier field appears on the CAN bus.
PRIO	Local priority This 3-bit field is only used when LPRIO_EN bit is set in MCR and it only makes sense for Tx buffers. These bits are not transmitted. They are appended to the regular ID to define the transmission priority. See Section 21.4.3 Arbitration process.
ID	Frame Identifier In Standard Frame format, only the 11 most significant bits (3 to 13) are used for frame identification in both receive and transmit cases. The 18 least significant bits are ignored. In Extended Frame format, all bits are used for frame identification in both receive and transmit cases.
DATA	Data Field As many as 8 bytes can be used for a data frame. For Rx frames, the data is stored as it is received from the CAN bus. For Tx frames, the CPU prepares the data field to be transmitted within the frame.

Table 399. Message buffer code for Rx buffers

Rx Code BEFORE Rx New Frame	Description	Rx Code AFTER Rx New Frame	Comment
0000	INACTIVE: MB is not active.	—	MB does not participate in the matching process.
0100	EMPTY: MB is active and empty.	0010	MB participates in the matching process. When a frame is received successfully, the code is automatically updated to FULL.
0010	FULL: MB is full.	0010	The act of reading the C/S word followed by unlocking the MB does not make the code return to EMPTY. It remains FULL. If a new frame is written to the MB after the C/S word was read and the MB was unlocked, the code still remains FULL.
		0110	If the MB is FULL and a new frame is overwritten to this MB before the CPU had time to read it, the code is automatically updated to OVERRUN. Refer to Section 21.4.5 Matching process for details about overrun behavior.

Table 399. Message buffer code for Rx buffers (continued)

Rx Code BEFORE Rx New Frame	Description	Rx Code AFTER Rx New Frame	Comment
0110	OVERRUN: a frame was overwritten into a full buffer.	0010	If the code indicates OVERRUN but the CPU reads the C/S word and then unlocks the MB, when a new frame is written to the MB the code returns to FULL.
		0110	If the code already indicates OVERRUN, and yet another new frame must be written, the MB will be overwritten again, and the code will remain OVERRUN. Refer to Section 21.4.5 Matching process for details about overrun behavior.
0XY1 ⁽¹⁾	BUSY: Flexcan is updating the contents of the MB. The CPU must not access the MB.	0010	An EMPTY buffer was written with a new frame (XY was 01).
		0110	A FULL/OVERRUN buffer was overwritten (XY was 11).

1. Note that for Tx MBs (see [Table 400](#)), the BUSY bit should be ignored upon read, except when AEN bit is set in the MCR.

Table 400. Message Buffer code for Tx buffers

RTR	Initial Tx code	Code after successful transmission	Description
X	1000	–	INACTIVE: MB does not participate in the arbitration process.
X	1001	–	ABORT: MB was configured as Tx and CPU aborted the transmission. This code is only valid when AEN bit in MCR is asserted. MB does not participate in the arbitration process.
0	1100	1000	Transmit data frame unconditionally once. After transmission, the MB automatically returns to the INACTIVE state.
1	1100	0100	Transmit remote frame unconditionally once. After transmission, the MB automatically becomes an Rx MB with the same ID.
0	1010	1010	Transmit a data frame whenever a remote request frame with the same ID is received. This MB participates simultaneously in both the matching and arbitration processes. The matching process compares the ID of the incoming remote request frame with the ID of the MB. If a match occurs this MB is allowed to participate in the current arbitration process and the Code field is automatically updated to '1110' to allow the MB to participate in future arbitration runs. When the frame is eventually transmitted successfully, the Code automatically returns to '1010' to restart the process again.
0	1110	1010	This is an intermediate code that is automatically written to the MB by the MBM as a result of match to a remote request frame. The data frame will be transmitted unconditionally once and then the code will automatically return to '1010'. The CPU can also write this code with the same effect.